

Nguyen Anh Khoi

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

About me

Backend Developer with a strong interest in building reliable, secure, and maintainable backend systems. I enjoy turning business requirements into practical software solutions, focusing on clear system design, clean implementation, and long-term scalability. Through academic projects, internship experience, and personal product development, I have developed a problem-solving mindset and the ability to learn quickly, work with complex workflows, and deliver backend features that support real user needs. reliability.

Technical Stack

Languages: Java, Go, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate/JPA, Gin, GORM, NestJS, FastAPI, Celery

Data: SQL Server, PostgreSQL, Redis, Qdrant, MinIO, Google Cloud Storage

Messaging/Streaming: RabbitMQ, gRPC, WebSocket, Server-Sent Events

Tools/Integrations: Docker, Docker Compose, Maven, Flyway, JWT, Playwright, VNPay, MoMo, Jsoup, BeautifulSoup

Experience

THD Cybersecurity

Jan 2026 – Jun 2026

Backend Dev, EMIS Voice Detection

Ho Chi Minh City, Vietnam

- Engineered a **NestJS recorder worker** that consumes RabbitMQ meeting jobs and automates Google Meet, Zoom, and Teams joins through Playwright-based browser automation.
- Scaled bot execution from **2 to 10 concurrent sessions** by decoupling meeting scheduling from recording workers and improving horizontal job orchestration.
- Reduced audio handoff latency from **3 minutes to 15 seconds** by streamlining recording upload, backend callbacks, and service-to-service processing flow.
- Integrated Go backend services with **gRPC streaming diarization**, WebSocket event broadcasting, Redis-backed streaming hubs, and MinIO-backed audio storage.
- Built **13 APIs/modules** across meeting lifecycle, targets, voice profiles, samples, chatbot, and history while supporting **2FA**, JWT, and cross-service APIs in a **21-service** system.

Technologies: Go, Gin, GORM, PostgreSQL, Redis, RabbitMQ, MinIO, gRPC, Docker, NestJS, Playwright, FastAPI, Celery, Qdrant

Projects

Carbon Credit Marketplace

Sep 2025 – Nov 2025

Backend Developer / GitHub Repository

- Owned the end-to-end **Spring Boot 3** backend for a carbon credit marketplace, delivering **120 REST APIs** across **18 modules** and **18 SQL Server entities**.
- Designed transaction-safe purchase flows with row-level reservation, idempotent completion, compensation rollback, wallet balance updates, audit logs, and buyer/seller notifications.
- Implemented stateless **JWT authentication**, token blacklisting, CORS, and method-level **RBAC** across **4 user roles** for marketplace, wallet, payment, and admin APIs.
- Engineered listing, transfer, dispute, retirement, certificate, and payment workflows, including VNPay routing, PDF certificate generation, and Google Cloud Storage upload support.
- Optimized marketplace and transaction query paths from **1437 ms to 24 ms**; added **30 requests/minute** rate limiting and Flyway-managed SQL indexes/constraints.

Technologies: Java 17, Spring Boot 3, Spring Security, Spring Data JPA, Hibernate/JPA, SQL Server, Flyway, JWT, Maven, Lombok, Thymeleaf, OpenHTMLtoPDF, Google Cloud Storage, VNPay, MoMo

Premier League Stats

Apr 2025 – Jul 2025

Backend Developer

- Built a football analytics backend that crawled real match data, normalized teams, fixtures, standings, and match statistics into **13 database entities**, and exposed **59 APIs**.
- Engineered proxy-forwarded scraping with request throttling, parser cleanup, retry handling, and data deduplication to reduce blocking risk during repeated ingestion jobs.
- Optimized crawler execution from **30 seconds to 4 seconds** by separating fetch, parse, transform, and persistence stages across a cleaner ingestion pipeline.
- Reduced memory usage from **17 MB to 493 KB** by minimizing retained objects, streaming parsed records, and avoiding oversized in-memory collections during scraping.
- Developed database-backed match analysis, formula-based prediction, role-protected APIs, and **VNPay/MoMo** payment workflows for payment-related operations.

Technologies: Java, Spring Boot, Spring Data JPA, Hibernate/JPA, SQL Server, Python, Jsoup, Requests, BeautifulSoup, JWT, RBAC, VNPay, MoMo

Education

FPT University

Sep 2023 – Expected Oct 2027

Bachelor of Software Engineering

GPA: 3.2/4.0